

Math Thermo
class # 24
4/23/2026

Recapitulation : Two-Phase Incompressible Flow

We have assumed that

- 1) $r = 2$ (binary mixture),
- 2) $\psi_i \equiv 0$, $i = 1, 2$ (no external potential),
- 3) $\sigma_k \equiv 0$, $k = 1, \dots, K$ (no reactions),
- 4) $p \equiv p_0 \equiv \text{constant}$ (incompressible).

We know that,

$$(24.1) \quad \nabla \cdot \vec{v} = 0,$$

$$(24.2) \quad \rho_0 \frac{\partial \phi_i}{\partial t} + \rho_0 \nabla \cdot (\phi_i \vec{v}) = -\nabla \cdot \vec{J}_i^D, \quad (i=1, 2)$$

$$(24.3) \quad \rho_0 \frac{\partial \vec{v}}{\partial t} + \rho_0 \nabla \cdot (\vec{v} \otimes \vec{v}) + \nabla p = -\nabla \cdot \Pi,$$

$$(24.4) \quad \frac{\partial \hat{u}}{\partial t} + \nabla \cdot (\hat{u} \vec{v}) = -\nabla \cdot \vec{J}_u^D - \Pi : \nabla \vec{v},$$

Our goal is to find all terms in the equation

$$\frac{\partial \hat{s}}{\partial t} + \nabla \cdot (\hat{s} \vec{v}) = -\nabla \cdot \vec{J}_s^D + \sigma_s.$$

specifically, $\vec{J}_s^D$ and σ_s . Here $\hat{s}$ is the entropy density and $\hat{u}$ is the internal energy density.

Under the assumptions that

$$\rho u =: \hat{u} = \tilde{u}(\tau) + \lambda_u \delta(\phi, \nabla \phi, \phi_2, \nabla \phi_2),$$

$$\rho s =: \hat{s} = \tilde{s}(\tau) + \lambda_s \delta(\phi, \nabla \phi, \phi_2, \nabla \phi_2),$$

where λ_u and λ_s are constants we have seen that, if $\vec{v}$ is the barycentric velocity,

$$\begin{aligned} (24.5) \quad \frac{\partial \hat{u}}{\partial t} + \nabla \cdot (\hat{u} \vec{v}) &= \frac{\partial \tilde{u}}{\partial \tau} \left(\frac{\partial \tau}{\partial t} + \nabla \cdot (\tau \vec{v}) \right) \\ &+ \lambda_u \omega \left(\frac{\partial \phi}{\partial t} + \nabla \cdot (\phi \vec{v}) \right) \\ &+ \lambda_u \varepsilon \nabla \cdot \left(\nabla \phi \frac{\partial \phi}{\partial t} \right) \\ &+ \lambda_u \varepsilon \left[\nabla \cdot (\nabla \phi \otimes \nabla \phi) \right] \cdot \vec{v}, \end{aligned}$$

$$\begin{aligned} (24.6) \quad \frac{\partial \hat{s}}{\partial t} + \nabla \cdot (\hat{s} \vec{v}) &= \frac{\partial \tilde{s}}{\partial \tau} \left(\frac{\partial \tau}{\partial t} + \nabla \cdot (\tau \vec{v}) \right) \\ &+ \lambda_s \omega \left(\frac{\partial \phi}{\partial t} + \nabla \cdot (\phi \vec{v}) \right) \\ &+ \lambda_s \varepsilon \nabla \cdot \left(\nabla \phi \frac{\partial \phi}{\partial t} \right) \\ &+ \lambda_s \varepsilon \left[\nabla \cdot (\nabla \phi \otimes \nabla \phi) \right] \cdot \vec{v}, \end{aligned}$$

where

$$\phi := \phi_2,$$

$$\omega := \frac{1}{\varepsilon} \tilde{W}'(\phi) - \varepsilon \Delta \phi, \quad \tilde{W}(\phi) := W(1-\phi, \phi).$$

Combining (24.4) and (24.5) we obtained the following heat equation:

$$\begin{aligned}
 (24.7) \quad \frac{\partial \tilde{u}}{\partial t} \left(\frac{\partial T}{\partial t} + \nabla \cdot (T \vec{v}) \right) &= -\nabla \cdot \vec{J}_v^0 - \Pi : \nabla \vec{v} \\
 &\quad - \lambda_u \omega \left(\frac{\partial \phi}{\partial t} + \nabla \cdot (\phi \vec{v}) \right) \\
 &\quad - \lambda_u \varepsilon \nabla \cdot \left(\nabla \phi \frac{\partial \phi}{\partial t} \right) \\
 &\quad - \lambda_u \varepsilon \left[\nabla \cdot (\nabla \phi \otimes \nabla \phi) \right] \cdot \vec{v}.
 \end{aligned}$$

Proposition (24.1): Suppose that thermal compatibility holds, ie,

$$(24.8) \quad \frac{\partial \tilde{S}}{\partial T} = \frac{1}{T} \frac{\partial \tilde{u}}{\partial T}$$

and assume that

$$(24.9) \quad p_u =: \hat{u}(T, \phi_1, \nabla \phi_1, \phi_2, \nabla \phi_2) = \tilde{u}(T) + \lambda_u \delta(\phi_1, \nabla \phi_1, \phi_2, \nabla \phi_2),$$

$$(24.10) \quad p_f =: \hat{f}(T, \phi_1, \nabla \phi_1, \phi_2, \nabla \phi_2) = \tilde{f}(T) + \lambda_f \delta(\phi_1, \nabla \phi_1, \phi_2, \nabla \phi_2),$$

$$(24.11) \quad p_s =: \hat{s}(T, \phi_1, \nabla \phi_1, \phi_2, \nabla \phi_2) = \tilde{s}(T) + \lambda_s \delta(\phi_1, \nabla \phi_1, \phi_2, \nabla \phi_2),$$

where λ_u and λ_s are constants and, further,

$$\tilde{f} = \tilde{u} - T \tilde{s} \quad \text{and} \quad \lambda_f = \lambda_u - T \lambda_s.$$

Then,

$$(24.12) \quad \frac{\partial \hat{S}}{\partial t} + \nabla \cdot (\hat{S} \vec{v}) = -\nabla \cdot \vec{J}_s^D + \sigma_s,$$

where

$$(24.13) \quad \vec{J}_s^D = \frac{1}{T} \vec{J}_v^D - \frac{\lambda_f}{\rho_0} \frac{\omega}{T} \vec{J}_2^D + \lambda_f \varepsilon \frac{1}{T} \nabla \phi \frac{\partial \phi}{\partial t} + \lambda_f \varepsilon \frac{1}{T} (\nabla \phi \otimes \nabla \phi) \vec{v},$$

$$(24.14) \quad \sigma_s = -\frac{1}{T} \nabla \vec{v} : (\Pi - \lambda_f \varepsilon \nabla \phi \otimes \nabla \phi) - \frac{\lambda_f}{\rho_0} \nabla \left(\frac{\omega}{T} \right) \cdot \vec{J}_2^D + \nabla \left(\frac{1}{T} \right) \cdot \left\{ \vec{J}_v^D + \lambda_u \varepsilon \nabla \phi \frac{\partial \phi}{\partial t} + \lambda_u \varepsilon (\nabla \phi \otimes \nabla \phi) \vec{v} - \frac{\lambda_s T \omega}{\rho_0} \vec{J}_2^D \right\}.$$

Proof: As indicated above

$$\begin{aligned} \frac{\partial \hat{S}}{\partial t} + \nabla \cdot (\hat{S} \vec{v}) &= \frac{\partial \tilde{S}}{\partial T} \left(\frac{\partial T}{\partial t} + \nabla \cdot (T \vec{v}) \right) \\ &+ \lambda_s \omega \left(\frac{\partial \phi}{\partial t} + \nabla \cdot (\phi \vec{v}) \right) \\ &+ \lambda_s \varepsilon \nabla \cdot \left(\nabla \phi \frac{\partial \phi}{\partial t} \right) \\ &+ \lambda_s \varepsilon \left[\nabla \cdot (\nabla \phi \otimes \nabla \phi) \right] \cdot \vec{v}, \end{aligned}$$

where

$$\omega := \frac{1}{\varepsilon} \tilde{W}'(\phi) - \varepsilon \Delta \phi, \quad \tilde{W}(\phi) := W(1-\phi, \phi)$$

and

$$\phi := \phi_2 \quad (\phi_1 = 1 - \phi).$$

Using thermal compatibility

$$\begin{aligned} \frac{\partial \hat{S}}{\partial t} + \nabla \cdot (\hat{S} \vec{v}) &= \frac{1}{T} \frac{\partial \tilde{u}}{\partial T} \left(\frac{\partial T}{\partial t} + \nabla \cdot (T \vec{v}) \right) \\ &+ \lambda_s \omega \left(\frac{\partial \phi}{\partial t} + \nabla \cdot (\phi \vec{v}) \right) \\ &+ \lambda_s \varepsilon \nabla \cdot \left(\nabla \phi \frac{\partial \phi}{\partial t} \right) \end{aligned}$$

$$+ \lambda_s \varepsilon \left[\nabla \cdot (\nabla \phi \otimes \nabla \phi) \right] \cdot \vec{v},$$

or, equivalently,

$$(24.15) \quad \begin{aligned} T \left(\frac{\partial \hat{s}}{\partial t} + \nabla \cdot (\hat{s} \vec{v}) \right) &= \frac{\partial \tilde{u}}{\partial T} \left(\frac{\partial T}{\partial t} + \nabla \cdot (T \vec{v}) \right) \\ &+ T \lambda_s \omega \left(\frac{\partial \phi}{\partial t} + \nabla \cdot (\phi \vec{v}) \right) \\ &+ T \lambda_s \varepsilon \nabla \cdot \left(\nabla \phi \frac{\partial \phi}{\partial t} \right) \\ &+ T \lambda_s \varepsilon \left[\nabla \cdot (\nabla \phi \otimes \nabla \phi) \right] \cdot \vec{v}. \end{aligned}$$

Combining

$$\frac{\partial \hat{u}}{\partial t} + \nabla \cdot (\hat{u} \vec{v}) = -\nabla \cdot \vec{J}_U^0 - \Pi : \nabla \vec{v}$$

and

$$\begin{aligned} \frac{\partial \hat{u}}{\partial t} + \nabla \cdot (\hat{u} \vec{v}) &= \frac{\partial \tilde{u}}{\partial T} \left(\frac{\partial T}{\partial t} + \nabla \cdot (T \vec{v}) \right) \\ &+ \lambda_u \omega \left(\frac{\partial \phi}{\partial t} + \nabla \cdot (\phi \vec{v}) \right) \\ &+ \lambda_u \varepsilon \nabla \cdot \left(\nabla \phi \frac{\partial \phi}{\partial t} \right) \\ &+ \lambda_u \varepsilon \left[\nabla \cdot (\nabla \phi \otimes \nabla \phi) \right] \cdot \vec{v} \end{aligned}$$

we recall the following form of the heat equation:

$$\begin{aligned}
 (24.16) \quad \frac{\partial \tilde{u}}{\partial t} \left(\frac{\partial T}{\partial t} + \nabla \cdot (T \vec{v}) \right) &= -\nabla \cdot \vec{J}_v^0 - \Pi : \nabla \vec{v} \\
 &\quad - \lambda_u \omega \left(\frac{\partial \phi}{\partial t} + \nabla \cdot (\phi \vec{v}) \right) \\
 &\quad - \lambda_u \varepsilon \nabla \cdot \left(\nabla \phi \frac{\partial \phi}{\partial t} \right) \\
 &\quad - \lambda_u \varepsilon \left[\nabla \cdot (\nabla \phi \otimes \nabla \phi) \right] \cdot \vec{v}.
 \end{aligned}$$

Substituting (24.16) into (24.15) we have

$$\begin{aligned}
 T \left(\frac{\partial \hat{S}}{\partial t} + \nabla \cdot (\hat{S} \vec{v}) \right) &= -\nabla \cdot \vec{J}_v^0 - \Pi : \nabla \vec{v} \\
 &\quad + (T \lambda_s - \lambda_u) \omega \left(\frac{\partial \phi}{\partial t} + \nabla \cdot (\phi \vec{v}) \right) \\
 &\quad + (T \lambda_s - \lambda_u) \varepsilon \nabla \cdot \left(\nabla \phi \frac{\partial \phi}{\partial t} \right) \\
 &\quad + (T \lambda_s - \lambda_u) \varepsilon \left[\nabla \cdot (\nabla \phi \otimes \nabla \phi) \right] \cdot \vec{v} \\
 &= -\nabla \cdot \vec{J}_v^0 - \Pi : \nabla \vec{v} \\
 &\quad - \lambda_f \omega \left(\frac{\partial \phi}{\partial t} + \nabla \cdot (\phi \vec{v}) \right) \\
 &\quad - \lambda_f \varepsilon \nabla \cdot \left(\nabla \phi \frac{\partial \phi}{\partial t} \right) \\
 &\quad - \lambda_f \varepsilon \left[\nabla \cdot (\nabla \phi \otimes \nabla \phi) \right] \cdot \vec{v}.
 \end{aligned}$$

Using the component mass conservation equation

$$\frac{\partial \phi}{\partial t} + \nabla \cdot (\phi \vec{v}) = - \frac{\nabla \cdot \vec{J}_z^0}{\rho_0},$$

we have

$$\begin{aligned}
 T \left(\frac{\partial \hat{S}}{\partial t} + \nabla \cdot (\hat{S} \vec{v}) \right) &= -\nabla \cdot \vec{J}_v^0 - \Pi : \nabla \vec{v} \\
 &\quad + \lambda_f \omega \frac{\nabla \cdot \vec{J}_2^0}{\rho_0} \\
 &\quad - \lambda_f \varepsilon \nabla \cdot \left(\nabla \phi \frac{\partial \phi}{\partial t} \right) \\
 &\quad - \lambda_f \varepsilon \left[\nabla \cdot (\nabla \phi \otimes \nabla \phi) \right] \cdot \vec{v}.
 \end{aligned}
 \tag{24.17}$$

Equivalently,

$$\begin{aligned}
 \frac{\partial \hat{S}}{\partial t} + \nabla \cdot (\hat{S} \vec{v}) &= \frac{-\nabla \cdot \vec{J}_v^0}{T} - \frac{-\Pi : \nabla \vec{v}}{T} \\
 &\quad + \frac{\lambda_f \omega}{\rho_0 T} \nabla \cdot \vec{J}_2^0 \\
 &\quad - \frac{\lambda_f \varepsilon}{T} \nabla \cdot \left(\nabla \phi \frac{\partial \phi}{\partial t} \right) \\
 &\quad - \frac{\lambda_f \varepsilon}{T} \left[\nabla \cdot (\nabla \phi \otimes \nabla \phi) \right] \cdot \vec{v}.
 \end{aligned}
 \tag{24.18}$$

Now, we use some identities:

$$\begin{aligned}
 \frac{\omega}{T} \nabla \cdot \vec{J}_2^0 &= \nabla \cdot \left(\frac{\omega}{T} \vec{J}_2^0 \right) - \nabla \left(\frac{\omega}{T} \right) \cdot \vec{J}_2^0, \\
 \frac{1}{T} \nabla \cdot \vec{J}_v^0 &= \nabla \cdot \left(\frac{1}{T} \vec{J}_v^0 \right) - \nabla \left(\frac{1}{T} \right) \cdot \vec{J}_v^0, \\
 \frac{1}{T} \nabla \cdot \left(\nabla \phi \frac{\partial \phi}{\partial t} \right) &= \nabla \cdot \left(\frac{1}{T} \nabla \phi \frac{\partial \phi}{\partial t} \right) - \nabla \left(\frac{1}{T} \right) \cdot \nabla \phi \frac{\partial \phi}{\partial t}, \\
 \left[\nabla \cdot (\nabla \phi \otimes \nabla \phi) \right] \cdot \vec{v} &= \nabla \cdot [(\nabla \phi \otimes \nabla \phi) \vec{v}] \\
 &\quad - (\nabla \phi \otimes \nabla \phi) : \nabla \vec{v}.
 \end{aligned}$$

Inserting into (24.18), we find

$$\begin{aligned}
 \frac{\partial \hat{S}}{\partial t} + \nabla \cdot (\hat{S} \vec{v}) &= -\nabla \cdot \left(\frac{1}{T} \vec{J}_0^D \right) + \nabla \left(\frac{1}{T} \right) \cdot \vec{J}_0^D \\
 &+ \frac{\lambda_f}{\rho_0} \nabla \cdot \left(\frac{\omega}{T} \vec{J}_2^D \right) - \frac{\lambda_f}{\rho_0} \nabla \left(\frac{\omega}{T} \right) \cdot \vec{J}_2^D \\
 &- \lambda_f \varepsilon \nabla \cdot \left(\frac{1}{T} \nabla \phi \frac{\partial \phi}{\partial t} \right) + \lambda_f \varepsilon \nabla \left(\frac{1}{T} \right) \cdot \nabla \phi \frac{\partial \phi}{\partial t} \\
 &- \frac{\lambda_f \varepsilon}{T} \nabla \cdot [(\nabla \phi \otimes \nabla \phi) \vec{v}] \\
 &+ \frac{\lambda_f \varepsilon}{T} (\nabla \phi \otimes \nabla \phi) : \nabla \vec{v} - \frac{\Pi : \nabla \vec{v}}{T}.
 \end{aligned}$$

One more identity should do it:

$$\begin{aligned}
 \frac{1}{T} \nabla \cdot [(\nabla \phi \otimes \nabla \phi) \vec{v}] &= \nabla \cdot \left[\frac{1}{T} (\nabla \phi \otimes \nabla \phi) \vec{v} \right] \\
 &- \nabla \left(\frac{1}{T} \right) \cdot [(\nabla \phi \otimes \nabla \phi) \vec{v}].
 \end{aligned}$$

Thus, we have

$$\begin{aligned}
 \frac{\partial \hat{S}}{\partial t} + \nabla \cdot (\hat{S} \vec{v}) &= -\nabla \cdot \left(\frac{1}{T} \vec{J}_0^D \right) + \nabla \left(\frac{1}{T} \right) \cdot \vec{J}_0^D \\
 &+ \frac{\lambda_f}{\rho_0} \nabla \cdot \left(\frac{\omega}{T} \vec{J}_2^D \right) - \frac{\lambda_f}{\rho_0} \nabla \left(\frac{\omega}{T} \right) \cdot \vec{J}_2^D \\
 &- \lambda_f \varepsilon \nabla \cdot \left(\frac{1}{T} \nabla \phi \frac{\partial \phi}{\partial t} \right) + \lambda_f \varepsilon \nabla \left(\frac{1}{T} \right) \cdot \nabla \phi \frac{\partial \phi}{\partial t} \\
 &+ \frac{\lambda_f \varepsilon}{T} (\nabla \phi \otimes \nabla \phi) : \nabla \vec{v} - \frac{\Pi : \nabla \vec{v}}{T} \\
 &- \lambda_f \varepsilon \nabla \cdot \left[\frac{1}{T} (\nabla \phi \otimes \nabla \phi) \vec{v} \right] \\
 &+ \lambda_f \varepsilon \nabla \left(\frac{1}{T} \right) \cdot [(\nabla \phi \otimes \nabla \phi) \vec{v}].
 \end{aligned}$$

(24.19)

Since

$$\lambda_f = \lambda_u - T \lambda_s,$$

where $\lambda_u, \lambda_s \in \mathbb{R}$, λ_f is not, in general, a constant and the purple terms above are not pure divergence terms.

We need to analyze three terms of the general form

$$-\lambda_f \varepsilon \nabla \cdot \left(\frac{1}{T} \vec{q} \right) = -\nabla \cdot \left(\frac{\lambda_u \varepsilon}{T} \vec{q} \right) + T \nabla \cdot \left(\frac{\lambda_s \varepsilon}{T} \vec{q} \right),$$

where $\vec{q}$ is a smooth vector field. Calculating, we can transform the last terms via the product rule:

$$\nabla \cdot \vec{q} = \nabla \cdot (T \frac{1}{T} \vec{q}) = \nabla T \cdot \left(\frac{1}{T} \vec{q} \right) + T \nabla \cdot \left(\frac{1}{T} \vec{q} \right).$$

Equivalently,

$$\begin{aligned} T \nabla \cdot \left(\frac{1}{T} \vec{q} \right) &= \nabla \cdot \vec{q} - \nabla T \cdot \left(\frac{1}{T} \vec{q} \right) \\ &= \nabla \cdot \vec{q} + \nabla \left(\frac{1}{T} \right) \cdot (T \vec{q}) \end{aligned}$$

Thus,

$$\begin{aligned} \mp \lambda_f \varepsilon \nabla \cdot \left(\frac{1}{T} \vec{q} \right) &= \mp \nabla \cdot \left(\frac{\lambda_u \varepsilon}{T} \vec{q} \right) \\ &\quad \pm \nabla \cdot (\lambda_s \varepsilon \vec{q}) \\ &\quad \pm \nabla \left(\frac{1}{T} \right) \cdot (\lambda_s T \varepsilon \vec{q}) \end{aligned}$$

$$\mp \lambda_f \varepsilon \nabla \cdot \left(\frac{1}{T} \vec{q} \right) = \mp \nabla \cdot \left(\frac{\lambda_f \varepsilon}{T} \vec{q} \right) \\ \pm \nabla \left(\frac{1}{T} \right) \cdot \left(\lambda_s T \varepsilon \vec{q} \right).$$

Therefore, transforming the purple terms using the last identity (24.19) becomes

$$\begin{aligned} \frac{\partial \hat{S}}{\partial t} + \nabla \cdot (\hat{S} \vec{v}) &= -\nabla \cdot \left(\frac{1}{T} \vec{J}_v^D \right) + \nabla \left(\frac{1}{T} \right) \cdot \vec{J}_v^D \\ &+ \nabla \cdot \left(\frac{\lambda_f \omega}{\rho_0 T} \vec{J}_2^D \right) - \frac{\lambda_f}{\rho_0} \nabla \left(\frac{\omega}{T} \right) \cdot \vec{J}_2^D \\ &- \nabla \cdot \left(\frac{\lambda_f \varepsilon}{T} \nabla \phi \frac{\partial \phi}{\partial t} \right) + \lambda_f \varepsilon \nabla \left(\frac{1}{T} \right) \cdot \nabla \phi \frac{\partial \phi}{\partial t} \\ &+ \frac{\lambda_f \varepsilon}{T} (\nabla \phi \otimes \nabla \phi) : \nabla \vec{v} - \frac{\Pi : \nabla \vec{v}}{T} \\ &- \nabla \cdot \left[\frac{\lambda_f \varepsilon}{T} (\nabla \phi \otimes \nabla \phi) \vec{v} \right] \\ &+ \lambda_f \varepsilon \nabla \left(\frac{1}{T} \right) \cdot [(\nabla \phi \otimes \nabla \phi) \vec{v}] \\ &- \nabla \left(\frac{1}{T} \right) \cdot \left(\frac{\lambda_s T \omega}{\rho_0} \vec{J}_2^D \right) \\ &+ \nabla \left(\frac{1}{T} \right) \cdot \left(\lambda_s T \varepsilon \nabla \phi \frac{\partial \phi}{\partial t} \right) \\ &+ \nabla \left(\frac{1}{T} \right) \cdot \left(\lambda_s T \varepsilon (\nabla \phi \otimes \nabla \phi) \vec{v} \right). \end{aligned} \tag{24.20}$$

Comparing (24.20) to (24.12), we find

$$\begin{aligned} \vec{J}_s^D &= \frac{1}{T} \vec{J}_v^D - \frac{\lambda_f}{\rho_0} \frac{\omega}{T} \vec{J}_2^D + \lambda_f \varepsilon \frac{1}{T} \nabla \phi \frac{\partial \phi}{\partial t} \\ &+ \lambda_f \varepsilon \frac{1}{T} (\nabla \phi \otimes \nabla \phi) \vec{v} \end{aligned}$$

$$\sigma_s = -\frac{1}{T} \nabla \vec{v} : (\Pi - \lambda_f \varepsilon \nabla \phi \otimes \nabla \phi) - \frac{\lambda_f}{\rho_0} \nabla \left(\frac{\omega}{T} \right) \cdot \vec{J}_2^D \\ + \nabla \left(\frac{1}{T} \right) \cdot \left\{ \vec{J}_v^D + \lambda_u \varepsilon \nabla \phi \frac{\partial \phi}{\partial t} + \lambda_u \varepsilon (\nabla \phi \otimes \nabla \phi) \vec{v} - \frac{\lambda_s T \omega}{\rho_0} \vec{J}_2^D \right\}.$$

Thus (24.13) and (24.14) are confirmed. ///

Other forms are possible. Consider the following identity:

$$\nabla \left(\frac{\omega}{T} \right) \cdot \vec{J}_2^D = \frac{1}{T} \nabla \omega \cdot \vec{J}_2^D + \nabla \left(\frac{1}{T} \right) \cdot (\omega \vec{J}_2^D).$$

Thus, we have the following:

Corollary (24.2): Under the same conditions as in the last proposition

$$\frac{\partial \hat{S}}{\partial t} + \nabla \cdot (\hat{S} \vec{v}) = -\nabla \cdot \vec{J}_s^D + \sigma_s,$$

where

$$(24.21) \quad \vec{J}_s^D = \frac{1}{T} \vec{J}_v^D - \frac{\lambda_f}{\rho_0} \frac{\omega}{T} \vec{J}_2^D + \lambda_f \varepsilon \frac{1}{T} \nabla \phi \frac{\partial \phi}{\partial t} \\ + \lambda_f \varepsilon \frac{1}{T} (\nabla \phi \otimes \nabla \phi) \vec{v}$$

and

$$(24.22) \quad \sigma_s = -\frac{1}{T} \nabla \vec{v} : (\Pi - \lambda_f \varepsilon \nabla \phi \otimes \nabla \phi) - \frac{\lambda_f}{\rho_0 T} \nabla \omega \cdot \vec{J}_2^D \\ + \nabla \left(\frac{1}{T} \right) \cdot \left\{ \vec{J}_0^D + \lambda_u \varepsilon \nabla \phi \frac{\partial \phi}{\partial t} + \lambda_u \varepsilon (\nabla \phi \otimes \nabla \phi) \vec{v} - \frac{\lambda_u}{\rho_0} \omega \vec{J}_2^D \right\}.$$

Constitutive Relations

Now, finally, we can choose Π , $\vec{J}_2^D$, and $\vec{J}_0^D$ in order to ensure that

$$\sigma_s \geq 0,$$

consistent with the second law of thermodynamics.

There are many ways to do this, as we shall see. For now, we take the simplest path forward.

Theorem (24.3): Under the assumptions of Proposition (24.1)

$$\sigma_s \geq 0$$

is guaranteed, provided

$$(24.23) \quad \Pi = -\mu \varepsilon + \lambda_f \varepsilon \nabla \phi \otimes \nabla \phi$$

where $\mu > 0$ and ε is the (divisive) shear rate tensor

$$\varepsilon := \frac{1}{2} (\nabla \vec{v} + (\nabla \vec{v})^T) - \frac{1}{3} \nabla \cdot \vec{v} \mathbb{I}_3 \\ = \frac{1}{2} (\nabla \vec{v} + (\nabla \vec{v})^T),$$

$$(24.24) \quad \vec{J}_2^D = -M_2 \nabla \omega,$$

where $M_2 > 0$ is a constant; and

$$(24.25) \quad \begin{aligned} \vec{J}_0^D = & M_T T^2 \nabla \left(\frac{1}{T} \right) - \lambda_u \varepsilon \nabla \phi \frac{\partial \phi}{\partial t} \\ & - \lambda_u \varepsilon (\nabla \phi \otimes \nabla \phi) \vec{v} - \frac{\lambda_u M_2}{\rho_0} \omega \nabla \omega, \end{aligned}$$

where $M_T > 0$ is a constant. In this case

$$(24.26) \quad \begin{aligned} \sigma_s = & \frac{\mu}{T} \nabla \vec{v} : \gamma + \frac{\lambda_f}{\rho_0 T} M_2 |\nabla \omega|^2 \\ & + M_T T^2 \left| \nabla \left(\frac{1}{T} \right) \right|^2 \\ \geq & 0. \end{aligned}$$

Proof: Substitute (24.23) – (24.25) into (24.22) to arrive at (24.26). Note that

$$\nabla \vec{v} : \gamma \geq 0$$

appealing to Proposition (22.2). //

Remark (24.4): Note that the positivity of the entropy production term relies on the fact the temperature function remains positive, well away from zero. This is a non-trivial assumption. Furthermore, we require $\lambda_f \geq 0$. Recall,

$$\lambda_f = \lambda_u - T \lambda_s.$$

The Complete Model

Now that the constitution relations have been chosen we can write the models in detail.

Let us assume that

$$\tilde{u} = p_0 C_h T,$$

$$\tilde{f} = p_0 C_h T - p_0 C_h T \ln\left(\frac{T}{T_0}\right),$$

$$\tilde{s} = p_0 C_h \ln\left(\frac{T}{T_0}\right).$$

As we have observed, thermal compatibility holds for this model

$$\frac{\partial \tilde{s}}{\partial T} = \frac{1}{T} \frac{\partial \tilde{u}}{\partial T}$$

and, clearly,

$$\tilde{f} = \tilde{u} - T \tilde{s}.$$

The important fact for the final form of our equations is that

$$\frac{\partial \tilde{u}}{\partial T} = p_0 C_h > 0$$

is a positive constant. C_h is known as the heat capacity.

The dynamical equations are, in all the details,

$$(24.27) \quad \nabla \cdot \vec{v} = 0, \quad (\text{Continuity})$$

$$(24.28) \quad \frac{\partial \phi}{\partial t} + \nabla \cdot (\phi \vec{v}) = \frac{M}{\rho_0} \Delta \omega, \quad \leftarrow (\text{Cahn-Hilliard})$$

$$(24.29) \quad \omega = \frac{1}{\epsilon} \tilde{W}'(\phi) - \epsilon \Delta \phi,$$

$$(24.30) \quad \rho_0 \left(\frac{\partial \vec{v}}{\partial t} + \nabla \cdot (\vec{v} \otimes \vec{v}) \right) + \nabla P = \nabla \cdot (\mu \gamma) - \lambda_f \epsilon \nabla \cdot (\nabla \phi \otimes \nabla \phi)$$

$$(24.31) \quad (\text{Navier-Stokes}) \quad \gamma = \frac{1}{2} (\nabla \vec{v} + (\nabla \vec{v})^T),$$

$$(24.32) \quad \lambda_f = \lambda_u - T \lambda_s,$$

$$(24.33) \quad \begin{aligned} \rho_0 C_h \left(\frac{\partial T}{\partial t} + \nabla \cdot (T \vec{v}) \right) = & M_+ \Delta T + \cancel{\lambda_u \epsilon \nabla \cdot \left(\nabla \phi \frac{\partial \phi}{\partial t} \right)} \\ & + \lambda_u \epsilon \nabla \cdot \left[(\nabla \phi \otimes \nabla \phi) \vec{v} \right] \\ & + \frac{\lambda_u M_2}{\rho_0} \nabla \cdot (\omega \nabla \omega) \\ & - \nabla \vec{v} : [\mu \gamma - \lambda_f \epsilon \nabla \phi \otimes \nabla \phi] \\ & - \lambda_u \omega \left(\frac{\partial \phi}{\partial t} + \nabla \cdot (\phi \vec{v}) \right) \\ & - \cancel{\lambda_u \epsilon \nabla \cdot \left(\nabla \phi \frac{\partial \phi}{\partial t} \right)} \\ & - \lambda_u \epsilon \left[\nabla \cdot (\nabla \phi \otimes \nabla \phi) \right] \cdot \vec{v}. \end{aligned}$$

(heat (Fourier) equation)

This is called the (or a)

Cahn-Hilliard-Navier-Stokes-Fourier

model.

Onsager - Casimir Relations

Theorem (24.5): Suppose that

$$\sigma_s = \sum_{i=1}^n J_i^\alpha X_i^\alpha + \sum_{i=1}^m J_i^\beta X_i^\beta,$$

where J_i^α, J_i^β are fluxes and X_i^α, X_i^β are even (α) and odd (β) thermodynamic forces. Assume that

$$J_i^\alpha = \sum_{j=1}^n L_{ij}^{\alpha,\alpha} X_j^\alpha + \sum_{j=1}^m L_{ij}^{\alpha,\beta} X_j^\beta, \quad 1 \leq i \leq n,$$

$$J_i^\beta = \sum_{j=1}^n L_{ij}^{\beta,\alpha} X_j^\alpha + \sum_{j=1}^m L_{ij}^{\beta,\beta} X_j^\beta, \quad 1 \leq i \leq m,$$

where

$$(24.34) \quad L_{ij}^{\alpha,\alpha} = L_{ji}^{\alpha,\alpha}, \quad 1 \leq i, j \leq n,$$

$$(24.35) \quad L_{ij}^{\beta,\beta} = L_{ji}^{\beta,\beta}, \quad 1 \leq i, j \leq m,$$

$$(24.36) \quad L_{ij}^{\alpha,\beta} = -L_{ji}^{\beta,\alpha}, \quad 1 \leq i \leq n, \quad 1 \leq j \leq m.$$

Set

$$L := \begin{bmatrix} L^{\alpha,\alpha} & L^{\alpha,\beta} \\ L^{\beta,\alpha} & L^{\beta,\beta} \end{bmatrix}, \quad \vec{J} := \begin{bmatrix} \vec{J}^\alpha \\ \vec{J}^\beta \end{bmatrix}, \quad \vec{X} := \begin{bmatrix} \vec{X}^\alpha \\ \vec{X}^\beta \end{bmatrix}.$$

Then

$$\sigma_s = X^T L_s X = \vec{X}^{\alpha T} L^{\alpha,\alpha} \vec{X}^\alpha + \vec{X}^{\beta T} L^{\beta,\beta} \vec{X}^\beta,$$

where

$$L_s = \frac{1}{2}(L + L^T) = \begin{bmatrix} L^{\alpha,\alpha} & 0 \\ 0 & L^{\beta,\beta} \end{bmatrix}.$$

Proof: Clearly,

$$\sigma_S = \vec{J}^T \vec{X}$$

and

$$\vec{J} = L \vec{X}.$$

In particular,

$$\vec{J} = L \vec{X} = \begin{bmatrix} L^{\alpha\alpha} \vec{X}^\alpha + L^{\alpha\beta} \vec{X}^\beta \\ L^{\beta\alpha} \vec{X}^\alpha + L^{\beta\beta} \vec{X}^\beta \end{bmatrix}.$$

Thus, since $L^{\alpha,\alpha^T} = L^{\alpha,\alpha}$ and $L^{\beta,\beta^T} = L^{\beta,\beta}$,

$$\vec{J}^T = \left[\vec{X}^{\alpha^T} L^{\alpha,\alpha} + \vec{X}^{\beta^T} L^{\alpha,\beta^T}, \vec{X}^{\alpha^T} L^{\beta,\alpha^T} + \vec{X}^{\beta^T} L^{\beta,\beta} \right]$$

and

$$\sigma_S = \vec{J}^T \vec{X}$$

$$= \vec{X}^{\alpha^T} L^{\alpha,\alpha} \vec{X}^\alpha + \vec{X}^{\beta^T} L^{\alpha,\beta^T} \vec{X}^\alpha$$

$$+ \vec{X}^{\alpha^T} L^{\beta,\alpha^T} \vec{X}^\beta + \vec{X}^{\beta^T} L^{\beta,\beta} \vec{X}^\beta$$

$$= \vec{X}^{\alpha^T} L^{\alpha,\alpha} \vec{X}^\alpha + \vec{X}^{\alpha^T} L^{\alpha,\beta} \vec{X}^\beta$$

$$+ \vec{X}^{\alpha^T} L^{\beta,\alpha} \vec{X}^\beta + \vec{X}^{\beta^T} L^{\beta,\beta} \vec{X}^\beta$$

$$= \vec{X}^{\alpha^T} L^{\alpha,\alpha} \vec{X}^\alpha + \vec{X}^{\alpha^T} L^{\alpha,\beta} \vec{X}^\beta$$

$$- \vec{X}^{\alpha^T} L^{\alpha,\beta} \vec{X}^\beta + \vec{X}^{\beta^T} L^{\beta,\beta} \vec{X}^\beta$$

$$= \vec{X}^{\alpha^T} L^{\alpha,\alpha} \vec{X}^\alpha + \vec{X}^{\beta^T} L^{\beta,\beta} \vec{X}^\beta,$$

Since

$$L^{\beta, \alpha^T} = -L^{\alpha, \beta}. \quad //$$

The relations (24.34) - (24.36) are called Onsager - Casimir relations.

Corollary (24.6): If $L^{\alpha, \alpha}$ and $L^{\beta, \beta}$ are positive semi-definite, then $\sigma_s \geq 0$.